# Uncertainty-Aware With Adaptive Geometric Correction for Multimodal Land-Cover Classification

Wenbin Cao, Xu Wang, Yi Xiao, *Member, IEEE*, Wenxin Huang, *Member, IEEE*, Bihan Wen, *Senior Member, IEEE*, and Xian Zhong, *Senior Member, IEEE*

***Abstract*—Land-cover classification (LCC) is a fundamental task in remote sensing and geographic information science. Multimodal fusion has shown great potential for enhancing LCC performance, for example, by combining optical and synthetic aperture radar (SAR) imagery to leverage their complementary strengths. However, two key challenges hinder effective fusion: 1) local geometric mismatches caused by distinct imaging geometries and 2) inconsistent reliability (the ability of a modality to deliver accurate and stable information) in LCC arising from different modalities and their acquisition conditions. To address these issues, we propose uncertainty-aware fusion with adaptive geometric (UAG) correction, which comprises three main components. First, the adaptive geometric correction module (AGCM) applies learnable pixel shifts to establish bidirectional local correlations between multiscale optical and SAR features, thereby mitigating spatial inconsistencies. Second, the adaptive uncertainty-aware dynamic fusion module (ADFM) employs evidential deep learning (EDL) to model uncertainty, defined as the extent of reliability deficiency, for each modality using the Dirichlet distribution and subjective logic, enabling confidence-aware feature weighting. Third, a lightweight multiscale decoder (MSD) integrates hierarchical features through a hybrid MLP–convolutional architecture, improving both segmentation efficiency and accuracy. We evaluate UAG on the WHU-OPT-SAR and DFC23 datasets, where experimental results demonstrate substantial improvements over state-of-the-art methods. The code will be released at https://github.com/cccwbin/UAGNet**

***Index Terms*—Geometric debiasing, land-cover classification (LCC), modality uncertainty, synthetic aperture radar (SAR)-optical multimodality.**

## I. Introduction

LAND-COVER classification (LCC) is a critical task in geoscience that categorizes the Earth's surface into regions such as farmlands, forests, water bodies, and urban areas using remote sensing imagery. LCC plays a pivotal role in environmental monitoring [1] and urban planning [2], providing essential insights for addressing global challenges including climate change, biodiversity conservation, and sustainable development.

As shown in Fig. 1(a), optical images capture abundant color and texture details, accurately reflecting the external characteristics of different land covers. However, under adverse conditions such as cloud occlusion, some regions may become blurred or invisible [3]. In contrast, synthetic aperture radar (SAR) imagery can penetrate clouds and rainfall, enabling all-weather imaging, but is often affected by speckle noise and geometric distortions (e.g., layover and foreshortening).

Existing methods [4], [5], [6], [7] exploit the complementary strengths of optical and SAR imagery to improve LCC performance. Nevertheless, they often overlook the geometric mismatch between modalities. Due to distinct imaging mechanisms and inherent noise, the same region may appear differently in optical and SAR data. As illustrated in Fig. 1(b), a building with clear geometric features in the optical image appears with blurred contours in the SAR image, leading to cross-modality noise contamination. To mitigate this, MoCG [8] employs graph convolutional networks for unidirectional geometric correction from optical to SAR imagery. However, as shown in Fig. 1(c), unidirectional correction can be insufficient in certain scenarios. Li et al. [9] proposed global alignment in horizontal and vertical directions but ignored local mismatches. Softformer [10] enhances local region perception within a single modality but lacks intermodal interaction. These observations highlight the importance of bidirectional local geometric correction.

In addition, modalities exhibit varying reliability, that is, the ability of a modality to deliver accurate and stable

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Wenbin Cao and Xian Zhong are with Sanya Science and Education Innovation Park, Wuhan University of Technology, Sanya 572025, China, and also with Hubei Key Laboratory of Transportation Internet of Things, School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan 430070, China (e-mail: 346747@whut.edu.cn; zhongx@whut.edu.cn).

Xu Wang is with the Hubei Key Laboratory of Transportation Internet of Things, School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan 430070, China, and also with the College of Information and Communication, National University of Defense Technology, Wuhan 430010, China (e-mail: wangxu9191@gmail.com).

Yi Xiao is with the School of Computer and Artificial Intelligence, Zhengzhou University, Zhengzhou 450001, China (e-mail: xy574475@gmail.com).

Wenxin Huang is with Hubei Key Laboratory of Big Data Intelligent Analysis and Application, School of Computer Science, Hubei University, Wuhan 430062, China (e-mail: wenxinhuang_wh@163.com).

Bihan Wen is with the Rapid-Rich Object Search Laboratory, School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore 639798 (e-mail: bihan.wen@ntu.edu.sg).

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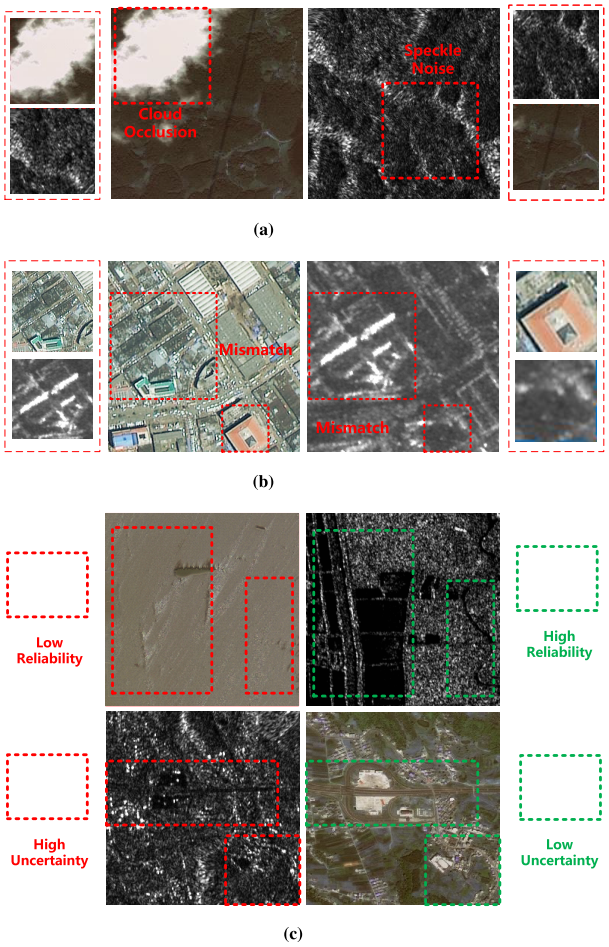



Fig. 1. Illustration of challenges in multimodal LCC. (a) Complementary characteristics of optical and SAR imagery. (b) Geometric mismatches between optical and SAR imagery. (c) Modality-dependent reliability variations and complementary uncertainty.

information under changing conditions, with uncertainty serving as its complementary measure by reflecting the extent of reliability deficiency. As shown in Fig. 1(c), this variability can significantly influence fusion quality. Current approaches fuse multimodal features using diverse strategies. For example, FTransUNet [7] applies direct addition for low-level features and cross-attention for high-level features; Li et al. [4] introduced gating mechanisms for selection and fusion; ExViT [11] leverages self-attention and token fusion; and CFFormer [12] combines cross-attention with MLP-based fusion. While effective, these methods do not explicitly evaluate modality reliability, which limits their potential for further performance improvements.

To address these challenges, we propose the uncertainty-aware fusion with adaptive geometric correction (UAG) network, which comprises three key components.

1) The adaptive geometric correction module (AGCM), illustrated in Fig. 2, applies learnable horizontal and vertical pixel shifts (HSL and VSL). Unlike patch-based methods, the AGCM adaptively shifts one modality's features relative to the other to capture fine-grained local spatial correlations, thereby achieving bidirectional correction. In addition, inspired by evidential deep learning (EDL), we introduce a variational loss to enhance semantic alignment between modalities.
2) The uncertainty-aware dynamic fusion module (ADFM) fuses optical and SAR features by weighting them according to their uncertainty. Following [13], we employ EDL with the Dirichlet distribution [14] to estimate pixel-level uncertainty for each modality. Regions with lower uncertainty, representing higher reliability, are assigned greater fusion weights, enabling uncertainty-aware feature integration.
3) A lightweight multiscale decoder (MSD) efficiently integrates multilevel features through a hybrid architecture [15], enhancing both segmentation accuracy and computational efficiency.

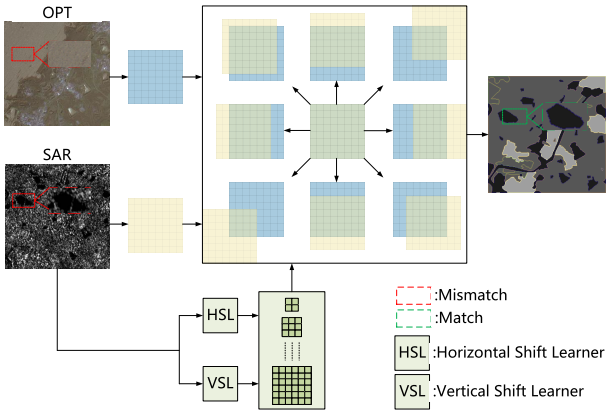



Fig. 2. Adaptive bidirectional local geometric correction. Example of SAR-guided correction: the optical feature map is fixed, while the SAR feature map is shifted with learnable offsets to adaptively capture geometric mismatches between optical and SAR images, thereby achieving geometric correction.

In summary, our contributions are threefold.

1) We propose the UAG network, which adaptively corrects geometric mismatches between optical and SAR imagery while explicitly evaluating modality reliability for accurate multimodal LCC.
2) We design an adaptive fine-grained geometric correction module using learnable pixel-level shifts to achieve bidirectional alignment between modalities.
3) We introduce an uncertainty-based adaptive fusion module that estimates modality uncertainty via the Dirichlet distribution and EDL, enabling confidence-aware fusion of multiscale features.

## II. Related Work

### A. Multimodal LCC

Multimodal LCC aims to improve classification performance by integrating the complementary characteristics of optical, SAR, and other data types. Numerous studies have explored novel strategies for multimodal feature fusion and network design. For instance, the CMFNet [16] employs cross-modal Transformers and multiscale context-enhanced modules to fuse optical imagery with DSM data. Other approaches [5], [17], [18] utilize techniques such as pseudo-twin feature extraction and multilevel cross-attention to enhance global

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context modeling and semantic consistency. Similarly, the CFNet [19] adopts cross-gated modules and multiscale fusion strategies to facilitate efficient interaction between optical and SAR features. Softformer [10] introduces an ISA module to enhance local information perception within a single modality, while ExViT [11] replaces linear projections with separable convolutions for tokenization and employs cross-attention mechanisms.

Several studies have also addressed modality discrepancies through distillation mechanisms. For example, DH-ADNet [20] and DAGDNet [21] employ collaborative instance selection and adaptive distillation learning to reduce semantic inconsistencies between modalities. MMOKD [22] and Assisted Learning [23] leverage knowledge distillation to further improve multimodal LCC performance.

Although these methods have demonstrated promising results, they often overlook local geometric mismatches between optical and SAR imagery. Such mismatches introduce spatial interference during cross-modal interaction, compromising the robustness of unimodal representations and ultimately degrading fusion performance. To address this limitation, we propose a bidirectional adaptive geometric correction module that employs pixel-level adaptive shift learning to achieve multiscale spatial alignment between optical and SAR images.

### B. Multimodal Fusion

Multimodal fusion is a critical step in LCC and can be broadly categorized into direct fusion, attention-based fusion, and gate-based fusion.

*1) Direct fusion* integrates multimodal features through simple operations such as addition or concatenation. For example, [24] combines heterogeneous visual features for image classification and RGB-D salient object detection. Yu et al. [25] fused inertial, EMG, visual, and environmental features for accurate real-time recognition of behaviors, expressions, and environments. Sun et al. [26] proposed LPEM, which encodes superpixel-level cross-modal structure and spatial label consistency into a single energy function for change detection. Similarly, SDIR [27] regresses optical and SAR data to form dual difference maps, builds superpixel similarity/dissimilarity graphs, and integrates them with an MRF to fuse global and local details.

*2) Attention-based fusion* dynamically weights features for enhanced integration. For instance, [28], [29] employ cross-attention mechanisms to fuse RGB with LiDAR or text, while [7], [30] apply attention modules for RGB-depth or RGB-hyperspectral fusion. He et al. [31] achieved unsupervised fusion of panchromatic and multispectral images via spectral domain transformation and adaptive feature fusion. Wang et al. [32] proposed a balanced multiview clustering method that integrates spectral clustering with K-means and adaptively weights each view. mmFormer [33] introduces a hybrid Transformer to model intramodal and intermodal dependencies, while CEN [34] uses adaptive channel exchange to balance shared and independent information.

*3) Gate-based fusion* filters redundant information to emphasize key features. Works such as [4], [35] incorporate gating mechanisms to dynamically adjust contributions from different modalities. More recent frameworks, including DynMM [36], MoE-Fusion [37], and ASANet [38], employ gating or cascade strategies to progressively integrate multimodal features, improving fusion effectiveness.

While each strategy has advantages, direct fusion is simple but less effective, and attention/gate-based methods are efficient but typically do not explicitly assess the reliability of each modality. In contrast, we propose an uncertainty-driven fusion approach based on the Dirichlet distribution and evidence theory. By explicitly modeling uncertainty, our method adaptively downweights unreliable modalities, thereby enhancing the robustness of multimodal feature representations.

### C. Uncertainty Estimation and Fusion

Uncertainty quantification interprets predictive bias arising from inherent sample noise as uncertainty and has been widely explored in machine learning. For example, TMC [39] employs subjective logic theory to model uncertainty in multiview images and combines evidential beliefs across perspectives. Jain et al. [40] adopted a bilevel optimization framework, using uncertainty scores in the inner loop to weight samples for outer-loop classification. Zhu et al. [41] tackled cross-domain crowd counting by modeling retrieval uncertainty via Dempster–Shafer evidence theory, while Liu et al. [13] addressed multiview action recognition by modeling prediction uncertainty with Dirichlet-based weighting in feature fusion. Gao et al. [42] modeled modality-specific uncertainty via multivariate Gaussian variance.

Uncertainty-based approaches have also been applied to remote sensing, particularly in multimodal fusion and semantic segmentation. Zheng et al. [43] used MC Dropout with Gaussian likelihood to estimate aleatoric and epistemic uncertainty in pansharpening. Shen et al. [44] and Wang et al. [45] leveraged entropy- and dropout-based measures to guide pseudo-labeling in semi-supervised segmentation and change detection. Li et al. [46] introduced a sigmoid-based probabilistic ranking to integrate uncertainty into feature fusion for building extraction, while Chen et al. [47] employed a Bayesian CNN with variational inference to decompose aleatoric and epistemic uncertainty in sea–ice classification.

Despite these advances, common estimators, including entropy-based scores [48], sampling methods such as MC Dropout [49] and deep ensembles [50], heteroscedastic regression [51], SNGP [52], energy-based scoring [53], and evidential learning, still face challenges for multimodal LCC. Entropy- and energy-based scores are efficient but provide only point confidence, often with inconsistent calibration across modalities or scales. Sampling-based methods (e.g., MC Dropout and deep ensembles) improve calibration but require costly forward passes. Heteroscedastic regression is lightweight, yet does not separate belief from uncertainty, while SNGP depends on feature geometry and is difficult to align across modalities. Most evidential approaches apply the Dirichlet distribution only at the output, leaving scale-wise reliability underexploited. Recent studies [42], [54], [55] incorporate uncertainty into fusion or alignment, but mainly

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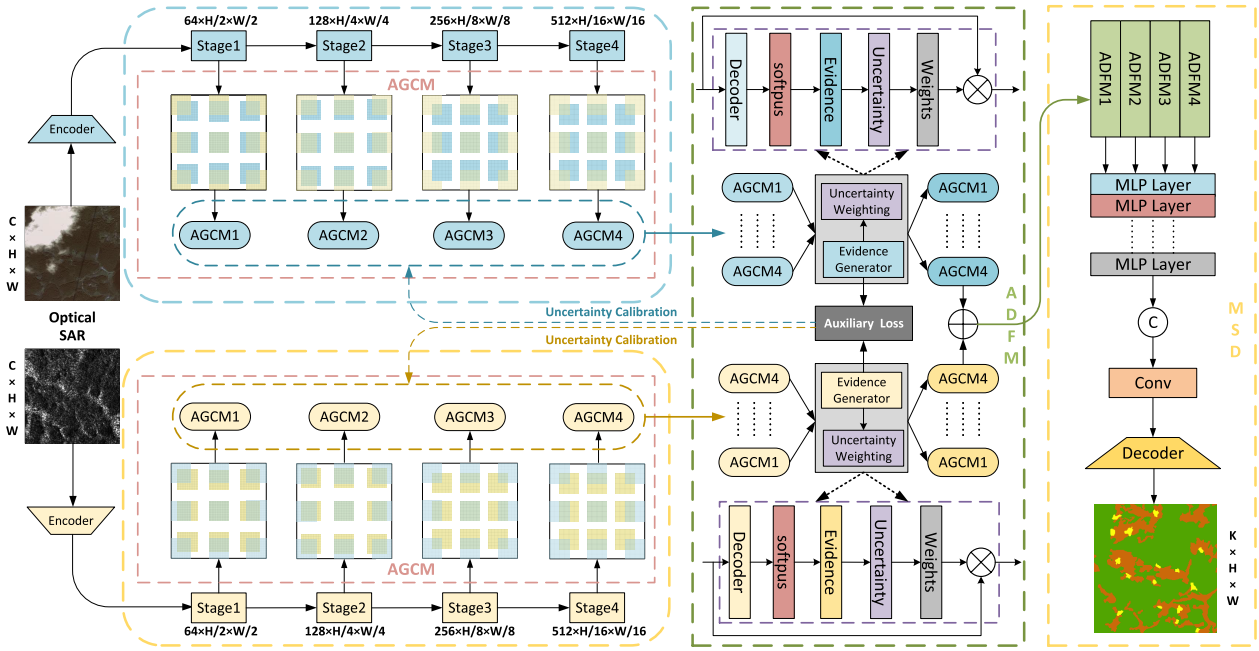



Fig. 3. Overall framework of the proposed UAG network. Optical and SAR features are first processed by the AGCM for bidirectional local geometric correction. The uncertainty-ADFM then models modality-specific uncertainty and dynamically fuses the multimodal features. Finally, the MSD aggregates the fused features to produce LCC results. $C$ denotes the number of channels, $H$ and $W$ represent the feature height and width, respectively, and $K$ indicates the number of classes.

at the decision level or a single scale and rarely provide a calibrated, interpretable signal consistent across modalities and resolutions.

To address these gaps, we propose an adaptive Dirichlet-based fusion module (ADFM) that predicts Dirichlet parameters at multiple feature scales, capturing scale-dependent reliability before decoding. The total evidence offers a unified confidence score that can be directly converted into adaptive fusion weights, avoiding complex evidence combination or costly sampling. This design enables stable, interpretable, and efficient integration of optical and SAR information, even in regions with strong cross-modal discrepancies, achieving a better balance between robustness and efficiency than existing uncertainty estimation and fusion approaches.

## III. Proposed Method

### A. Overall Framework

To fully exploit the complementarity between optical and SAR images while effectively addressing geometric mismatches, we propose the UAG network for multimodal LCC. The network not only compensates for geometric discrepancies between modalities but also dynamically adjusts their uncertainty during feature fusion.

As shown in Fig. 3, the framework consists of three components: the AGCM, the uncertainty-ADFM, and the MSD. The AGCM performs bidirectional adaptive geometric correction by computing pixel-level local correlations between optical and SAR features, thereby mitigating performance degradation caused by mismatches. The ADFM fuses features based on the Dirichlet distribution and subjective logic theory, dynamically weighting modalities according to reliability and suppressing uncertain information. Finally, the MSD integrates multiscale information through MLP layers and convolutional decoders to produce accurate LCC maps.

### B. Adaptive Geometric Correction Module

Geometric mismatch between optical and SAR imagery arises from their fundamentally different imaging mechanisms (see Fig. 1). Existing studies have addressed this challenge from several perspectives:1) explicit registration methods [56], [57] that correct pixel- or region-level distortions; 2) feature-level alignment strategies [38], [58], [59] that employ learnable transformations such as deformable convolutions; and 3) robust fusion or sensitivity analyses [17], [60] designed to mitigate the impact of residual misregistration on downstream tasks.

Our AGCM targets learnable geometric misalignments at the feature level between optical and SAR feature maps. By introducing pixel-wise displacements and bidirectional correlations, AGCM performs local adaptive correction of cross-modal representations, thereby improving feature consistency and enhancing land-cover segmentation under imperfect alignment. Importantly, AGCM is not intended to enforce complete geometric equivalence between modalities. Instead, it alleviates locally correctable geometric distortions, those most detrimental to classification accuracy, thereby complementing global registration and other alignment techniques.

We adopt ResNet50 [61] as the backbone to extract multiscale features from both optical and SAR images. To enhance channel awareness, we insert a lightweight squeeze-and-excitation (SE) module [62] between ResNet layers

$$G_{M_i}^{k} = \mathrm{Conv}_r\left(X_{M_i}\right) + S_k\left(\mathrm{Conv}_r\left(X_{M_i}\right)\right), \quad i \in \{1, 2\} \tag{1}$$

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where $G^k_{M_i}$ denotes the output of modality $M_i$ ($M_1$: optical, $M_2$: SAR) at the $k$th scale. $\text{Conv}_r$ indicates ResNet blocks with downsampling rates $\{2, 4, 8, 16\}$ and $S_k$ denotes the SE operation at scale $k$.

To achieve local geometric alignment, we define a horizontal shift learner (HSL) and a vertical shift learner (VSL) that adaptively estimate pixel-level shifts between modalities. For optical-guided SAR correction, the SAR feature map is fixed, while the optical feature map is iteratively shifted by learned step sizes. Local correlations are computed at each shift, concatenated with the original feature map, and convolved to obtain corrected features

$$\text{Cor}(x, y) = \mathcal{C}(\text{cor}(x, y, \delta m, \delta n)) \tag{2}$$

$$\delta^k_{M_2} = \mathcal{C}\left(\text{Cor}\left(G^k_{M_2}, G^k_{M_1}\right), G^k_{M_2}\right) \tag{3}$$

$$\delta^k_{M_2} = \text{Conv}\left(\delta^k_{M_2}\right), \quad k \in \{1, 2, 3, 4\} \tag{4}$$

where $\mathcal{C}$ denotes concatenation and Conv is a $3\times3$ convolution. The pixel-level correlation is defined as

$$\text{cor}(x, y, \delta m, \delta n) = x \cdot \mathcal{S}_y(\delta m, \delta n) \tag{5}$$

$$\mathcal{S}_y(\delta m, \delta n) = \mathcal{S}(y, \delta m, \delta n) \tag{6}$$

where $\mathcal{S}$ represents the shift operation applied to $y$ with stride $(\delta m, \delta n)$.

Shift strides are sampled from a learnable set

$$m, n \in s \sim \mathcal{U}(-a, a, N) \tag{7}$$

where $s$ is the learnable set of candidate pixel shifts, $\mathcal{U}$ denotes uniform sampling, $a$ is the maximum shift range (in pixels), and $N$ is the number of candidates. Here, $m$ and $n$ denote the selected *horizontal* and *vertical* pixel shifts, respectively. Strides are then normalized as follows:

$$\delta m = \frac{m}{W}, \quad \delta n = \frac{n}{H} \tag{8}$$

with $W$ and $H$ denoting feature width and height, respectively. The final output $\delta^k_{\text{SAR}}$ represents the corrected SAR features at scale $k$.

### C. Uncertainty-Aware Dynamic Fusion Module

Feature fusion is central to multimodal LCC. While direct addition or concatenation is simple, it ignores modality reliability. Attention- or gate-based methods improve adaptivity but lack interpretable uncertainty modeling.

We propose an uncertainty-ADFM that combines the Dirichlet distribution with subjective logic. The Dirichlet parameters represent evidence for each class, and the Dirichlet strength quantifies modality confidence. Subjective logic decomposes this into belief and uncertainty, yielding an interpretable reliability measure.

Specifically, features $\delta^k_{M_i}$ from each modality are first decoded by a lightweight shared decoder. Evidence values are obtained by applying a Softplus function with scaling factor $t$

$$d^k_{M_i} = \text{Softplus}\left(\text{Dec}\left(\delta^k_{M_i}\right) \cdot t\right) + 1. \tag{9}$$

The Dirichlet strength and corresponding uncertainty are defined as follows:

$$S^k_{M_i} = \sum_{c=1}^{C} d^k_{M_i,c}, \quad u^k_{M_i} = \frac{C}{S^k_{M_i}} \tag{10}$$

where $C$ is the number of classes. Higher $S^k_{M_i}$ indicates lower uncertainty and higher reliability.

Fusion weights are computed as follows:

$$\lambda^k_{M_i} = \frac{1 - u^k_{M_i}}{2 - \sum_{i=1}^{n} u^k_{M_i}} \tag{11}$$

normalized by

$$\lambda^k_{M_i} \leftarrow \frac{\lambda^k_{M_i}}{\sum_{i=1}^{n} \lambda^k_{M_i}}. \tag{12}$$

The fused feature at scale $k$ is then

$$f^k = \lambda^k_{M_1} \cdot x^k_1 + \lambda^k_{M_2} \cdot x^k_2. \tag{13}$$

### D. Multiscale Segmentation Decoder

To integrate hierarchical features, we design a lightweight MSD that combines MLPs and convolutions. For each scale $k$

$$\hat{f}^k = \text{Up}\left(\text{MLP}\left(f^k\right)\right) \tag{14}$$

$$\hat{y} = \mathcal{P}\left(\text{Conv}\left(\mathcal{C}\left(\hat{f}^k\right)\right)\right) \tag{15}$$

where Up is upsampling, $\mathcal{C}$ denotes concatenation, and $\mathcal{P}$ is the final convolutional decoder.

### E. Loss Function

The total loss of our model consists of two components: a segmentation loss and an auxiliary loss.

*1) Segmentation Loss:* The motivation for our segmentation loss design arises from the observation that minority classes (e.g., *road* and *others* in WHU-OPT-SAR) contribute little to the overall loss and are easily dominated by majority classes (e.g., *forest* and *farmland*). To mitigate this issue, we adopt the Dice loss, which is particularly suitable for imbalanced data, and further incorporate a category-based dynamic weighting strategy to strengthen the contribution of minority classes.

Class weights are computed based on the pixel proportion of each class. Specifically, the weight for class $i$ is defined as follows:

$$w_i = \frac{1}{\text{class_ratio}_i} \tag{16}$$

where $\text{class_ratio}_i$ denotes the proportion of pixels belonging to class $i$. The segmentation loss is then expressed as follows:

$$\mathcal{L}_{\text{seg}} = 1 - \frac{2\sum_i w_i (y_i \cdot \hat{y}_i)}{\sum_i w_i (y_i + \hat{y}_i)} \tag{17}$$

where $y_i$ and $\hat{y}_i$ represent the ground-truth and predicted masks for class $i$, respectively. This formulation effectively alleviates class imbalance and emphasizes difficult regions, thereby improving the robustness and accuracy of LCC.

*2) Auxiliary Loss:* The AGCM module enforces spatial alignment between modalities but does not guarantee semantic consistency. To address this, the ADFM incorporates an EDL framework based on the Dirichlet distribution and subjective logic. We, therefore, introduce a variational loss that naturally consists of two terms: a cross-entropy data-fitting component and a KL divergence regularizer. This design aligns modality-specific predictions with the ground-truth labels while constraining overconfident evidence assignment,

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thus enhancing semantic consistency and ensuring reliable uncertainty modeling.

The cross-entropy loss is defined as follows:

$$\mathcal{L}_{\mathrm{CE}}(d, y) = -\frac{1}{N_{\mathrm{pix}}} \sum_{\mathrm{pix}} \sum_{i=1}^{K} y_i \log p_i \tag{18}$$

where

$$p_i = \frac{d_i}{S_i} \tag{19}$$

where $d_i$ denotes the Dirichlet parameter for class $i$, $S_i = \sum_{j=1}^{K} d_j$ is the Dirichlet strength, $y_i$ is the one-hot ground-truth label, $K$ is the number of classes, and $N_{\mathrm{pix}}$ is the total number of pixels. Averaging over two modalities and $N$ scales yields

$$\mathcal{L}_{\mathrm{cls}} = \frac{1}{2N} \sum_{k=1}^{N} \left[ \mathcal{L}_{\mathrm{CE}}\left(d_{M_1}^{(k)}, y\right) + \mathcal{L}_{\mathrm{CE}}\left(d_{M_2}^{(k)}, y\right) \right]. \tag{20}$$

The KL divergence regularization term is formulated as follows:

$$\begin{aligned} \mathcal{L}_{\mathrm{KL}}\left(\mathrm{Dir}(d) \,\|\, \mathrm{Dir}(1)\right) &= \ln B(1) - \ln B(d) \\ &+ \sum_{c=1}^{K} (d_c - 1) \left[ \psi(d_c) - \psi\left(\textstyle\sum_{i=1}^{K} d_i\right) \right] \end{aligned} \tag{21}$$

where $B(\cdot)$ is the beta function and $\psi(\cdot)$ is the digamma function. The averaged KL regularization loss is

$$\begin{aligned} \mathcal{L}_{\mathrm{reg}} = \frac{1}{2N} \sum_{k=1}^{N} \Bigg[ & \mathcal{L}_{\mathrm{KL}}\left(\mathrm{Dir}\left(d_{M_1}^{(k)}\right) \Big\| \mathrm{Dir}(1)\right) \\ & + \mathcal{L}_{\mathrm{KL}}\left(\mathrm{Dir}\left(d_{M_2}^{(k)}\right) \Big\| \mathrm{Dir}(1)\right) \Bigg]. \end{aligned} \tag{22}$$

The auxiliary loss is finally defined as follows:

$$\mathcal{L}_{\mathrm{aux}} = \mathcal{L}_{\mathrm{cls}} + \alpha \cdot \mathcal{L}_{\mathrm{reg}} \tag{23}$$

where $\alpha = 0.1$ balances the KL regularization term.

*3) Total Loss:* The total training loss is the weighted sum of the segmentation loss and the auxiliary loss

$$\mathcal{L}_{\mathrm{total}} = \mathcal{L}_{\mathrm{seg}} + \beta \cdot \mathcal{L}_{\mathrm{aux}} \tag{24}$$

where $\beta$ controls the contribution of the auxiliary loss.

## IV. Experimental Results

### A. Dataset and Implementation Details

*1) Dataset Description:* WHU-OPT-SAR [5] is a large-scale multimodal benchmark for LCC, containing 100 optical–SAR image pairs (5-m resolution, 5556 × 3704 pixels) annotated with seven land-use categories: farmland, city, village, water, forest, road, and others. Following official guidelines, images are cropped into 256 × 256 patches and split into 17 640/5880/5880 patches for training, validation, and testing.

DFC23 [63], from the IEEE GRSS 2023 Data Fusion Contest, provides optical–SAR pairs for roof-type classification across 12 geometry-based categories. We follow the official protocol, converting the instance segmentation task into 12-class semantic segmentation and splitting the data into training, validation, and testing sets at a 7:2:1 ratio.

In our work, we follow the official settings and convert the original 12-category instance segmentation task into a 12-category semantic segmentation task. The dataset is randomly divided into training, validation, and testing subsets with a 7:2:1 ratio.

*2) Implementation Details:*

*a) Baseline model:* The baseline model adopts a U-Net-like encoder–decoder architecture [64]. In the encoding phase, we employ a ResNet50 backbone pretrained on ImageNet [65], incorporating a dual-stream structure for optical and SAR branches. A feature pyramid structure is applied for multiscale feature fusion, and the multimodal features are fused by direct addition. In the decoding phase, a U-Net-style decoder generates the final LCC maps.

*b) Training setting:* All experiments were conducted on a single NVIDIA RTX 4090 GPU. Training was performed for 300 epochs on both datasets. The batch size was set to 8 for WHU-OPT-SAR and 4 for DFC23. We used the Adam optimizer with an initial learning rate of $10^{-4}$ and set $\alpha = 0.1$ and $\beta = 1$. Early stopping monitored validation loss, halting training if no improvement was observed within 10 epochs to prevent overfitting. Gradient clipping was also applied to avoid gradient explosion and stabilize training.

*3) Evaluation Metrics:* To comprehensively evaluate segmentation performance, we adopted five widely used metrics: mean intersection over union (mIoU), overall accuracy (OA), the Kappa coefficient, mean $F1$-score (mF1), and user's accuracy (UA).

The mIoU quantifies the average ratio of intersection to union between the predicted and ground-truth masks across all classes. OA measures the proportion of correctly classified pixels over all pixels. The Kappa coefficient evaluates inter-rater agreement by comparing OA with the accuracy expected by random chance, providing a more objective measure for imbalanced datasets. The mean $F1$-score (mF1) represents the harmonic mean of precision and recall across all classes, while UA reflects the proportion of correctly classified pixels within each predicted class, highlighting commission errors. Together, these metrics provide a balanced and reliable assessment of model performance.

### B. Experimental Results on WHU-OPT-SAR

*1) Comparative Analysis With Multimodal Methods:* We compared our proposed UAG network with existing multimodal methods on WHU-OPT-SAR test set (see Table I). TransFuser [66] uses a Transformer-based mid-fusion strategy, achieving an OA of 80.3% and an mIoU of 52.4%. The CMFNet [16] and DDHRNet [17], which employ cross-modal and cross-scale attention Transformers, obtain OAs of 81.1% and 82.0%, and mIoUs of 52.1% and 52.3%, respectively. In contrast, the CFNet [19], based on a gating mechanism, achieves a lower OA of 79.9%, highlighting its limited ability to exploit complementary features.

Mmformer [33] adopts a dual-Transformer strategy, achieving an OA of 83.8% and an mIoU of 55.1%. MCANet

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TABLE I
EXPERIMENTAL RESULTS (%) OF THE PROPOSED UAG MODEL COMPARED WITH REPRESENTATIVE METHODS ON THE WHU-OPT-SAR DATASET

| Method | Venue | UA per category | | | | | | | OA | Kappa | mF1 | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Farmland | City | Village | Water | Forest | Road | Others | | | | |
| TransFuser [66] | CVPR'21 | 81.3 | 75.3 | 68.3 | 83.4 | 87.6 | 58.6 | 56.0 | 82.6 | 75.3 | 70.3 | 56.2 |
| CMFNet [16] | JSTARS'22 | 80.0 | 66.7 | 64.3 | 79.7 | **90.8** | 54.3 | 24.7 | 81.1 | 73.4 | 66.5 | 52.1 |
| MCANet [5] | JAG'22 | 82.9 | 72.4 | 62.8 | 78.0 | 89.9 | 42.3 | 20.1 | 81.7 | 74.0 | 66.5 | 52.3 |
| DDHRNet [17] | JAG'22 | 84.2 | 69.6 | 59.7 | 80.1 | 89.9 | 48.1 | 23.8 | 82.0 | 74.4 | 66.6 | 52.3 |
| CFNet [19] | JSTARS'22 | 81.5 | 80.9 | 50.5 | 68.8 | 90.5 | 19.9 | 37.5 | 79.9 | 71.3 | 66.2 | 52.7 |
| mmFormer [33] | MICCAI'22 | **84.8** | **81.6** | **72.8** | 83.5 | 87.8 | 67.9 | 53.2 | 83.8 | 77.2 | 67.0 | 55.1 |
| CEN [34] | TPAMI'22 | 84.5 | 74.1 | 66.5 | 79.8 | 89.4 | 52.1 | 24.8 | 82.8 | 75.6 | 70.1 | 56.2 |
| TokenFusion [67] | CVPR'22 | 84.2 | 78.5 | 65.6 | 81.5 | 90.3 | 55.8 | 31.4 | 83.6 | 76.8 | 72.2 | 58.5 |
| ExViT [11] | TGRS'23 | 81.0 | 72.6 | 69.9 | 82.1 | 88.7 | 59.1 | 51.6 | 82.7 | 75.4 | 69.4 | 55.5 |
| CFFormer [12] | TGRS'24 | 81.7 | 70.9 | 70.0 | 82.8 | 88.5 | 62.0 | 56.5 | 82.9 | 75.8 | 69.7 | 55.9 |
| MultiSenseSeg [68] | TGRS'24 | 70.1 | 74.9 | 68.5 | 81.5 | 88.7 | 58.6 | 54.8 | 83.0 | 75.9 | 70.5 | 56.6 |
| FTransUNet [7] | TGRS'24 | 81.0 | 72.5 | 71.0 | 81.6 | 90.0 | 60.7 | 52.5 | 83.0 | 76.0 | 70.8 | 56.9 |
| MFNet [69] | TGRS'25 | 69.0 | 58.4 | 47.8 | 63.4 | 83.9 | 39.5 | 20.3 | 84.0 | 51.9 | 68.5 | 54.6 |
| PAD [70] | TGRS'25 | 69.9 | 58.1 | 49.7 | 65.4 | 84.1 | 43.2 | 23.5 | **84.6** | 53.9 | 70.1 | 56.3 |
| UAG (Ours) | | 83.6 | 76.7 | 70.4 | **84.5** | 89.7 | **69.2** | **60.5** | **84.6** | **78.2** | **74.1** | **60.6** |

TABLE II
ABLATION EXPERIMENT RESULTS (%) OF THE MAIN MODULES ON THE WHU-OPT-SAR AND DFC23 DATASETS

| Modules | | | | WHU-OPT-SAR | | | | DFC23 | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Baseline | MSD | AGCM | ADFM | mIoU | OA | Kappa | mF1 | mIoU | OA | Kappa | mF1 |
| ● | ○ | ○ | ○ | 58.7 | 83.7 | 76.8 | 72.5 | 54.0 | 93.8 | 80.0 | 77.2 |
| ● | ● | ○ | ○ | 59.0 | 83.8 | 77.0 | 72.7 | 65.5 | 94.7 | 83.2 | 78.5 |
| ● | ○ | ● | ○ | 59.2 | 83.7 | 76.9 | 73.0 | 60.6 | 94.0 | 81.2 | 78.6 |
| ● | ○ | ○ | ● | 58.9 | 83.8 | 77.0 | 72.6 | 54.9 | 94.0 | 83.9 | 79.7 |
| ● | ● | ● | ○ | 60.0 | 84.2 | 77.6 | 73.6 | 67.2 | 94.9 | 81.2 | 78.1 |
| ● | ● | ○ | ● | 59.5 | 84.0 | 77.4 | 73.2 | 72.3 | 95.2 | 85.2 | 83.5 |
| ● | ○ | ● | ● | 59.6 | 84.1 | 77.4 | 73.2 | 63.7 | 94.9 | 84.2 | 81.1 |
| ● | ● | ● | ● | **60.6** | **84.6** | **78.2** | **74.1** | **78.0** | **96.5** | **89.0** | **87.4** |

[5] leverages pseudo-Siamese feature extraction and cross-attention, achieving an OA of 81.7% and an mIoU of 58.3%. TokenFusion [67] integrates features via dynamic token replacement and positional alignment, obtaining an OA of 83.6% and an mIoU of 58.5%. ExViT [11] proposes a lightweight Transformer, achieving an OA of 82.7% and an mIoU of 55.5%. CEN [34] employs channel swapping for adaptive feature exchange, reaching an OA of 82.8% and an mIoU of 56.2%. MultiSenseSeg [68] and FTransUNet [7] further enhance intermodal interaction, achieving mIoUs of 56.6% and 56.9%.

In comparison, our UAG network incorporates adaptive bidirectional geometric correction and uncertainty-aware feature fusion. To address class imbalance, we also adopt a class-based dynamic weighted loss. As shown in Table I, this strategy significantly improves UA for minority classes while slightly affecting others. Overall, UAG achieves substantial gains in both OA and mIoU, attaining state-of-the-art performance.

*2) Ablation Studies:*

*a) Module ablation:* Ablation studies on WHU-OPT-SAR further demonstrate the contribution of each proposed module. As shown in Table II, incorporating the AGCM significantly improves both mIoU and OA compared to configurations without it. The AGCM leverages complementary cues from optical and SAR data, using optical color and texture to reduce SAR distortions, and SAR contours to refine object boundaries.

TABLE III
COMPARISON OF PERFORMANCE (%) ON THE WHU-OPT-SAR DATASET USING DIFFERENT LOSS FUNCTIONS

| seg_loss | cdw | aux_loss | mIoU | OA | Kappa | mF1 |
|---|---|---|---|---|---|---|
| ce | ○ | ○ | 58.9 | 84.2 | 77.5 | 72.5 |
| ce | ● | ○ | 59.2 | 83.6 | 77.6 | 72.8 |
| dice | ○ | ○ | 59.8 | 84.2 | 77.7 | 73.4 |
| dice | ● | ○ | 60.3 | 84.4 | 78.1 | 73.9 |
| dice | ● | ● | **60.6** | **84.6** | **78.2** | **74.1** |
| dice + ce | ○ | ○ | 59.5 | 84.2 | 77.6 | 73.0 |
| dice + ce | ● | ○ | 59.5 | 83.9 | 77.4 | 73.2 |
| dice + ce | ● | ● | 59.8 | 84.0 | 77.5 | 73.4 |

Adding the ADFM further improves all metrics by dynamically weighting modalities based on reliability. Replacing the conventional CNN-based decoder with MSD also enhances performance, confirming its strength in aggregating multiscale features.

When modules are combined, the AGCM and ADFM together outperform the baseline and individual modules, highlighting their necessity. Performance gains are even more pronounced when MSD is included, as the AGCM/ADFM refines multiscale features while MSD aggregates them efficiently. The final results demonstrate that all three modules complement one another, fully validating the UAG network design.

*b) Loss ablation and hyperparameters:* As shown in Table III, the comparison between rows 1 and 2 as well as

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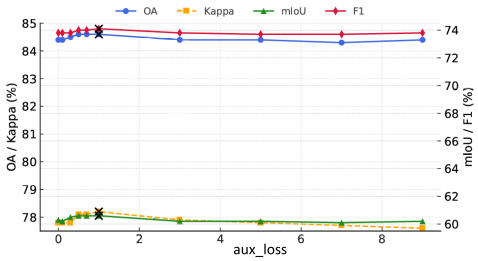



Fig. 4. Comparison of performance (%) on the WHU-OPT-SAR dataset with different values of the `aux_loss` hyperparameter.

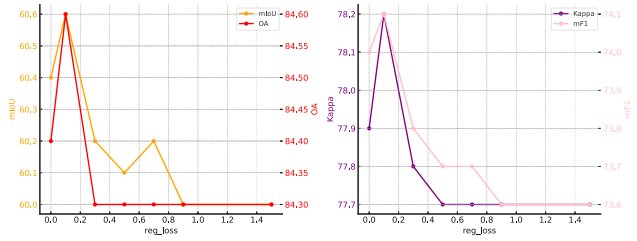



Fig. 5. Comparison of performance (%) on the WHU-OPT-SAR dataset with different values of the `reg_loss` hyperparameter.

3 and 4 confirms the effectiveness of the class-wise dynamic weighting (cdw) mechanism. Examining rows 1/3/6 and 2/4/7 further reveals that combining CE and Dice losses yields worse performance than using Dice loss alone, while both weighted and unweighted Dice consistently outperform CE.

This suggests that Dice loss is better suited to class-imbalanced remote sensing segmentation, as it directly optimizes region overlap (IoU), is more sensitive to small targets and boundaries, and aligns more closely with evaluation metrics. In contrast, CE's pixel-wise formulation and background dominance lead to small-target neglect. Using CE together with Dice may also introduce conflicting optimization objectives, global (CE) versus local (Dice), which destabilizes training.

During training, we jointly optimized `seg_loss` and `aux_loss`, where `aux_loss` is composed of `cls_loss` and `reg_loss`. To identify optimal hyperparameters, we conducted ablation experiments on both `aux_loss` and `reg_loss`. As shown in Figs. 4 and 5, the best performance across all metrics was achieved when the `aux_loss` coefficient was set to 1 and the `reg_loss` coefficient to 0.1.Moreover, experiments over a reasonably broad range of `aux_loss` values indicate that this loss is largely insensitive to its hyperparameter, which can be attributed to its role as semantic guidance for intermediate representations rather than a dominant optimization term.To reduce randomness, each setting was run three times, and the results were averaged.

*c) Fusion strategies:* We further compared uncertainty-aware dynamic fusion strategies (u-early, u-mid, and u-late) with traditional fusion methods such as addition, concatenation, gating, and attention (see Table IV). The u-mid strategy significantly outperformed all others, achieving top IoUs of 72.3%, 62.7%, 54.1%, 72.9%, and 82.3% across the five most challenging categories, while maintaining competitive performance in the remaining ones. These results validate the

TABLE IV
COMPARISON OF DIFFERENT FUSION METHODS ON THE WHU-OPT-SAR DATASET (IOU, %)

| Method | IoU (%) | | | | | | |
|---|---|---|---|---|---|---|---|
| | 1 | 2 | 3 | 4 | 5 | 6 | 7 |
| Addition | 71.7 | 62.3 | 53.8 | 72.0 | 82.0 | **49.7** | 30.1 |
| Concatenation | 71.4 | 62.2 | 53.6 | 72.0 | 81.7 | 49.6 | 29.8 |
| Gating | 71.7 | 62.0 | 53.9 | 72.1 | 81.9 | 49.7 | 29.8 |
| Attention | 71.9 | 62.3 | 53.7 | 72.2 | 82.0 | 49.4 | **30.5** |
| UAG (u-early) | 71.8 | 62.2 | 53.5 | 72.2 | 81.9 | 48.6 | 28.7 |
| UAG (u-mid) | **72.3** | **62.7** | **54.1** | **72.9** | **82.3** | 49.6 | 30.3 |
| UAG (u-late) | 72.1 | 62.1 | 53.6 | 72.4 | 82.0 | 49.2 | 30.2 |

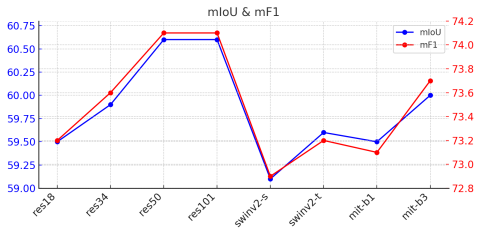



Fig. 6. Comparison of results on the WHU-OPT-SAR dataset using different backbone networks.

superiority of uncertainty-aware dynamic fusion in multimodal LCC.

*d) Backbone ablation:* To ensure a fair comparison between different backbone architectures, we evaluated various backbones on the WHU-OPT-SAR dataset (see Fig. 6). For CNN architectures, we employed different configurations of the ResNet network. Results show that increasing network depth enhances feature representation, leading to consistent improvements in performance. However, performance gains plateau between ResNet50 and ResNet101. We attribute this to the larger parameter size of ResNet101, which may cause overfitting and reduce generalization ability.

For Transformer architectures, we selected Swin Transformer V2 [71] and Mix Transformer [15], both of which support multiscale feature extraction. Experimental results indicate that Transformer backbones underperform relative to CNN-based counterparts. These findings validate our choice of CNN architectures for feature extraction in remote sensing imagery and highlight their advantage in capturing discriminative features.

*e) Uncertainty estimation methods:* We compared our approach against several representative uncertainty estimation methods, including entropy-based weighting, MC dropout, deep ensembles, SNGP, and heteroscedastic regression, as summarized in Table V.

For multimodal LCC, traditional measures such as entropy-based [48] and heteroscedastic regression [51] primarily rely on point estimates of predictive variance and often underestimate uncertainty in regions with strong cross-modal discrepancies. MC dropout [49] and deep ensembles [50] improve calibration through sampling or model averaging but require multiple forward passes and tend to smooth fine-scale boundaries, leading to suboptimal mIoU. SNGP [52] leverages spectral normalization and Gaussian processes for improved

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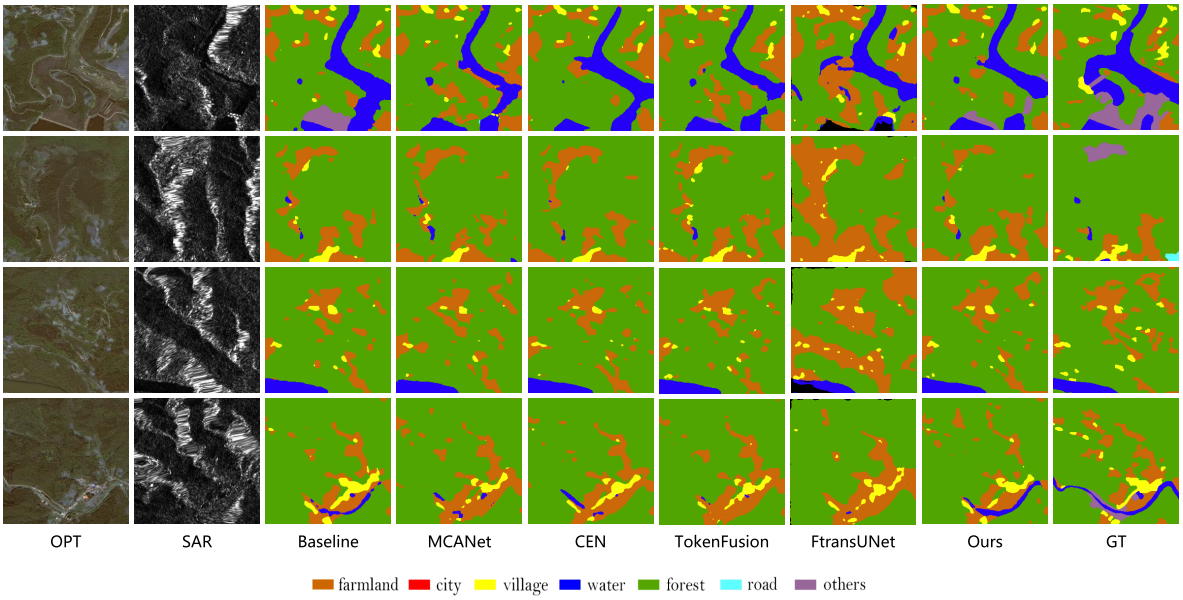



Fig. 7. Visualization of prediction results from the proposed UAG model, the baseline, and comparative methods on WHU-OPT-SAR test set.

TABLE V
COMPARISON OF DIFFERENT UNCERTAINTY ESTIMATION METHODS ON WHU-OPT-SAR

| UE Method | OA (%) | mIoU (%) | mF1 (%) | Kappa (%) |
|---|---|---|---|---|
| Entropy-based [48] | 84.3 | 60.2 | 73.8 | 77.7 |
| MC Dropout [49] | 84.4 | 60.4 | 74.0 | 77.8 |
| Deep Ensemble [50] | 84.4 | 60.3 | 73.9 | 77.8 |
| Heteroscedastic [51] | 84.3 | 60.2 | 73.8 | 77.7 |
| SNGP [52] | 84.2 | 59.9 | 73.5 | 77.6 |
| Energy-based [53] | 84.4 | 60.4 | 74.0 | 77.9 |
| EDL [39] | 84.4 | 60.3 | 73.9 | 77.8 |
| UAG (Ours) | **84.6** | **60.6** | **74.1** | **78.2** |

TABLE VI
PARAMETER SIZE AND PERFORMANCE COMPARISON OF DIFFERENT DECODER ARCHITECTURES

| Decoder Architecture | Decoder Params (M) | Total Params (M) | Total FLOPs (B) | mIoU (%) | OA (%) |
|---|---|---|---|---|---|
| UNet | 1.28 | 65.94 | 79.75 | **60.6** | 84.5 |
| ASPP | 2.26 | 66.92 | 76.02 | 59.9 | 84.2 |
| PSP | 0.48 | 65.14 | **73.57** | 58.7 | 83.4 |
| UperNet | 1.32 | 65.98 | 86.87 | **60.6** | 84.5 |
| UAG (Ours) | **0.30** | **64.96** | 75.88 | **60.6** | **84.6** |

out-of-distribution detection, yet its dense segmentation performance is constrained by the Gaussian process approximation. Energy-based uncertainty estimation [53] offers a sampling-free alternative but provides limited gains for fine-scale class separation, EDL [39] infers subjective evidence to model uncertainty, yet its direct evidence aggregation is insufficient to resolve cross-modal reliability variation.

As shown in Table V, all baselines achieve OA in the range of 84.2%–84.4% and mIoU below 60.5%. In contrast, our evidential learning strategy, using the Dirichlet distribution and subjective logic, achieves the best overall performance (OA = 84.6%, mIoU = 60.6%, mF1 = 74.1%, and Kappa = 78.2%). These results demonstrate that modeling evidence, rather than relying solely on sampling or variance-based heuristics, provides a richer and more interpretable uncertainty representation. This enables more accurate fusion of optical and SAR features and consistently better segmentation performance.

*f) Decoder parameter and efficiency comparison:* As summarized in Table VI, our SegFormer-style decoder maintains a compact design, with only 0.30 M parameters and one of the lowest computational costs (75.88 B FLOPs). Despite being substantially lighter than the ASPP-style (2.26 M) and UperNet-style (1.32 M) decoders, it achieves competitive mIoU (60.6%) and the highest OA (84.6%). These results suggest that our design achieves an effective tradeoff between segmentation accuracy and computational efficiency at the decoder level, rather than merely emphasizing lightweightness.

*3) Visualization Analysis:* Visualization results in Fig. 7 further highlight the advantages of UAG. Our method maintains sharper boundaries and reduces misclassifications in complex regions such as water, roads, and forests. The baseline frequently confuses water with roads and farmland. While MCANet [5] and CEN [34] offer partial improvements, they still misclassify water and urban-farmland regions. TokenFusion [67] and FTransUNet [7] perform worse, exhibiting significant blending between categories. Overall, UAG produces more accurate, fine-grained, and consistent segmentation results.

## C. Experimental Results on DFC23

*1) Comparative Analysis With State-of-the-Art Methods:* On DFC23, our proposed UAG method achieved the best results with an OA of 96.5% and an mIoU of 78.0% (see Table VII). FTransUNet [7], MultiSenseSeg [68], and TransFuser [66] followed with OAs of 96.3%, 95.8%, and 95.7%, respectively, but their mIoUs were 6.9%, 6.3%, and 8.5% lower than ours.

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TABLE VII
EXPERIMENTAL RESULTS (%) OF THE PROPOSED UAG MODEL AND COMPARATIVE METHODS ON THE DFC23 DATASET

| Method | Venue | IoU per category | | | | | | | | | | | | | OA | mIoU |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | BG | FR | GR | GR2 | RR | MER | HRv1 | HRv2 | MR | PR | AR | Do | OR | | |
| TransFuser [66] | CVPR'21 | 96.4 | 77.3 | 70.7 | 71.2 | 86.5 | 68.3 | 46.5 | 62.7 | 62.0 | 56.0 | 71.0 | **86.0** | 68.9 | 95.7 | 71.1 |
| CEN [34] | TPAMI'22 | 93.5 | 56.9 | 41.7 | 42.6 | 53.7 | 2.0 | 5.9 | 24.6 | 12.2 | 8.9 | 21.2 | 14.2 | 17.3 | 91.1 | 30.4 |
| mmFormer [33] | MICCAI'22 | 91.7 | 67.7 | 58.8 | 72.2 | 76.3 | 10.5 | 37.4 | 42.0 | 12.0 | 5.0 | 46.0 | 37.4 | 44.8 | 90.9 | 41.2 |
| CFNet [19] | JSTARS'22 | 84.2 | 44.7 | 22.8 | 31.2 | 41.1 | 0.0 | 0.0 | 6.8 | 0.0 | 0.0 | 7.8 | 2.8 | 19.8 | 83.9 | 20.5 |
| CMFNet [16] | JSTARS'22 | 96.5 | 77.1 | 71.8 | 72.6 | 82.6 | 69.2 | 46.9 | 64.7 | 40.9 | 57.3 | 63.6 | 70.9 | 64.1 | 95.7 | 67.5 |
| MCANet [5] | JAG'22 | 94.3 | 76.6 | 68.8 | 70.4 | 71.4 | 45.1 | 46.5 | 61.5 | 69.6 | 57.8 | 73.1 | 78.9 | 67.8 | 94.2 | 67.9 |
| TokenFusion [67] | CVPR'22 | 96.3 | 76.4 | 70.3 | 70.7 | 86.2 | 64.6 | 40.6 | 61.9 | 53.6 | 52.8 | 70.3 | 82.9 | 65.8 | 95.6 | 68.6 |
| ExViT [11] | TGRS'23 | 94.4 | 62.8 | 51.8 | 52.2 | 63.9 | 31.1 | 18.0 | 32.4 | 24.7 | 22.3 | 42.8 | 50.5 | 40.3 | 92.5 | 45.2 |
| SoftFormer [10] | ISPRS'24 | 90.7 | 48.2 | 33.9 | 34.4 | 42.6 | 0.2 | 0.0 | 14.4 | 0.1 | 0.0 | 4.1 | 0.0 | 19.9 | 88.7 | 22.2 |
| CFFormer [12] | TGRS'24 | 89.0 | 38.5 | 30.6 | 27.0 | 33.8 | 6.8 | 3.9 | 13.0 | 0.6 | 8.0 | 15.8 | 46.2 | 14.2 | 87.2 | 25.3 |
| MultiSenseSeg [68] | TGRS'24 | 96.5 | 78.4 | 69.8 | 73.4 | 86.1 | 62.0 | 47.8 | 56.8 | 64.8 | 55.8 | 66.5 | 77.1 | 68.7 | 95.8 | 69.5 |
| FTransUNet [7] | TGRS'24 | **97.1** | 79.8 | 74.2 | 72.4 | 86.0 | 72.8 | 48.7 | 66.5 | 59.8 | 59.8 | 70.0 | 77.5 | 67.2 | 96.3 | 71.7 |
| UAG (Ours) | | 96.8 | **80.5** | **76.5** | **76.4** | **90.1** | **75.4** | **62.5** | **74.2** | **78.8** | **71.6** | **73.8** | 83.2 | **74.7** | **96.5** | **78.0** |

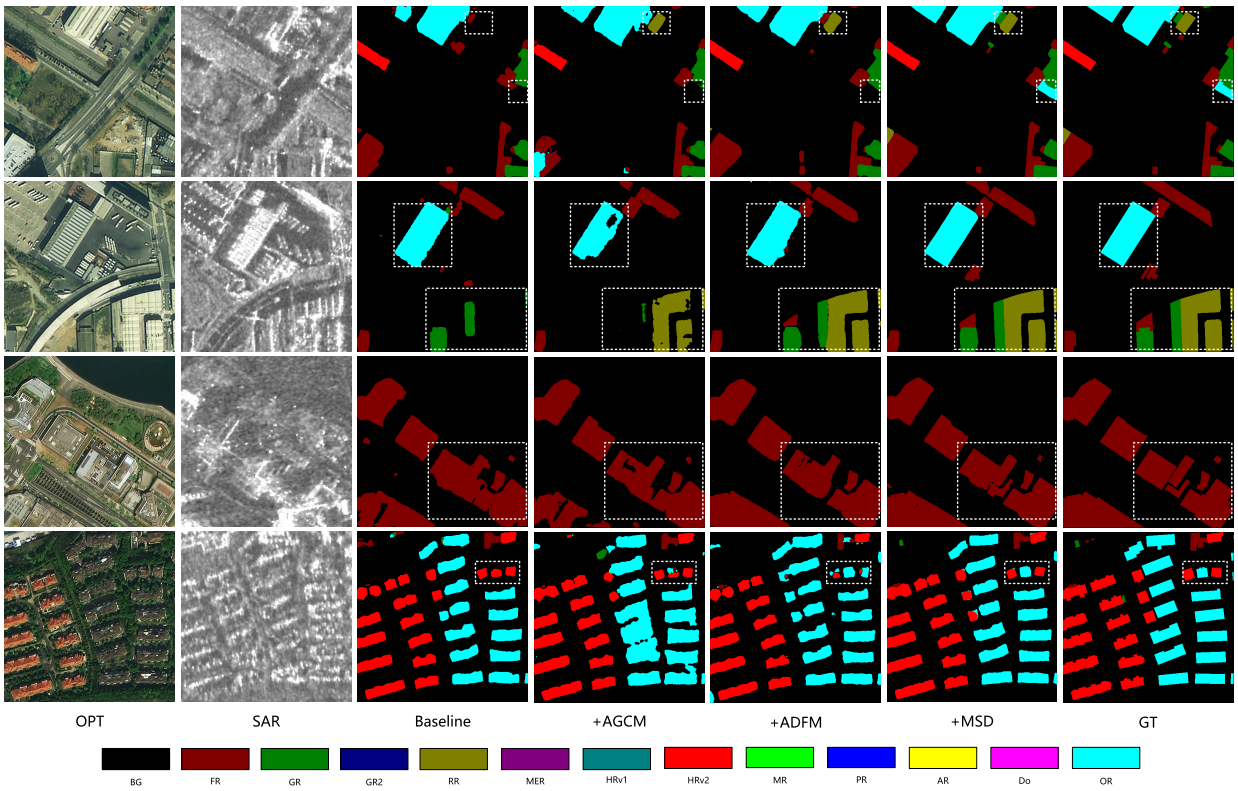



Fig. 8. Ablation study visualization on DFC23 test set, highlighting the contribution of each component.

We attribute this gap to three factors: 1) Transformer-based methods often lose local details compared to CNN-based networks; 2) these methods lack explicit geometric correction, leading to feature misalignment; and 3) they do not explicitly assess modality reliability, allowing weaker modalities to dominate fusion. In contrast, our method integrates adaptive geometric correction and uncertainty-aware fusion based on EDL, leading to superior accuracy and generalizability.

*2) Ablation Studies:*

*a) Quantitative analysis:* As shown in Table II, the AGCM improves mIoU from 54.0 to 60.6 by enhancing local geometric consistency between modalities. The ADFM raises the Kappa coefficient from 80.0 to 83.9, confirming its effectiveness in reliability-aware fusion. Replacing the U-Net-style decoder with MSD further improves all metrics, validating its efficiency in multiscale feature aggregation.

Pairwise ablations also confirm module complementarity: AGCM + ADFM outperforms the baseline and individual modules, and integrating MSD achieves the best performance. These results demonstrate that combining the AGCM, ADFM, and MSD fully leverages their strengths, validating the UAG network design.

*3) Module Ablation:* As illustrated in Fig. 8, the baseline fails to capture fine building details due to insufficient geometric alignment. Incorporating the AGCM provides fine-grained corrections, recovering more building structures. However, without uncertainty modeling, residual errors remain. The ADFM addresses this by weighting modalities according to

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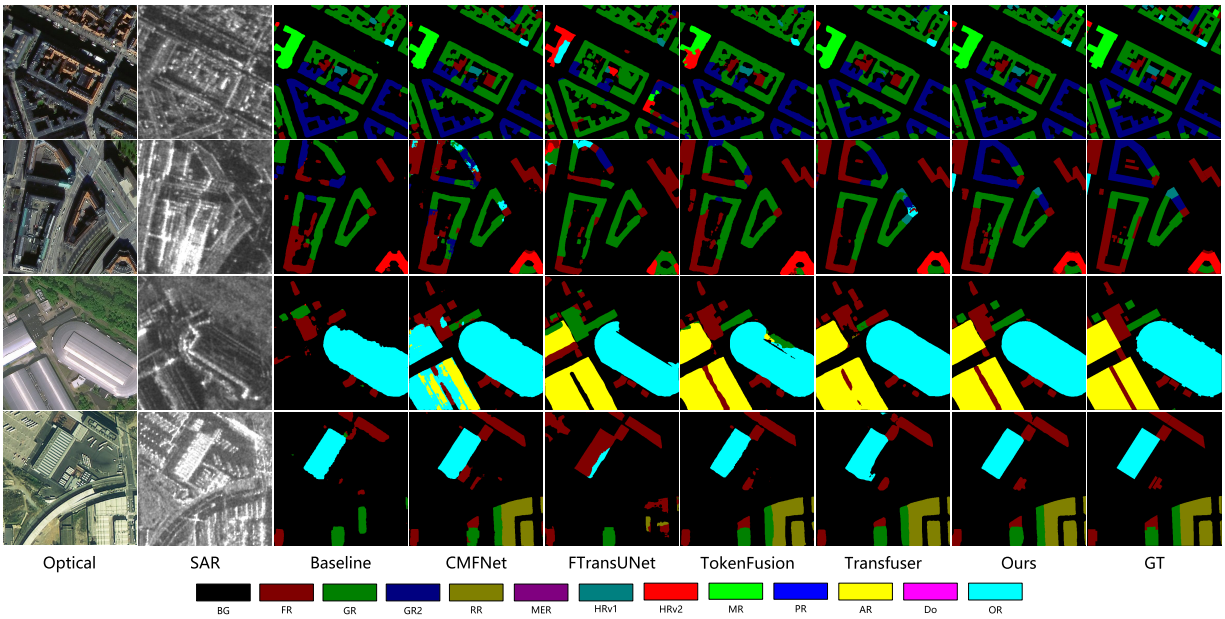



Fig. 9. Visualization of prediction results from the proposed UAG model, the baseline, and comparative methods on the DFC23 test set.

reliability, improving both boundary and semantic accuracy. Replacing the U-Net decoder with MSD further enhances performance by better integrating multiscale features. These ablation visualizations confirm the complementary contributions of the AGCM, ADFM, and MSD, which together significantly improve multimodal LCC.

*4) Visualization Analysis:* Visualization comparisons (see Fig. 9) highlight differences among methods. The baseline and CEN [34] exhibit major omissions and misclassifications in complex areas. TokenFusion [67] performs better but still struggles with accuracy. The CMFNet [16] and TransFuser [66], which combine CNN and Transformer features, reduce some misclassifications but remain prone to errors. By contrast, UAG effectively mitigates spatial and semantic misalignment through adaptive geometric correction, variational semantic alignment, and uncertainty-aware fusion, producing the most reliable classification results.

## V. Conclusion

In this article, we proposed an uncertainty-aware fusion framework with adaptive geometric correction to address geometric misalignment and varying reliability between optical and SAR images. Specifically, we introduced the AGCM, which performs bidirectional geometric correction through adaptive shift learning, and the uncertainty-aware dynamic fusion module (ADFM), which enables more robust multimodal feature fusion. In addition, we developed a lightweight MSD that efficiently aggregates features across multiple resolutions. Experimental results on WHU-OPT-SAR and DFC23 datasets demonstrate that our method substantially improves classification accuracy in multimodal LCC tasks.

For future work, we plan to explore more refined geometric correction mechanisms to better address complex modality differences and further enhance multimodal feature alignment. We also aim to investigate alternative theoretical frameworks for uncertainty estimation to guide dynamic fusion, to improve performance in diverse and challenging LCC scenarios.

## Acknowledgment

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**Wenbin Cao** received the B.S. degree in tourism management from Zhengzhou University, Zhengzhou, China, in 2021. He is currently pursuing the M.S. degree with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan, China.

His research interests include multimodal remote sensing image processing.



**Xu Wang** received the B.S. degree in software engineering from the School of Software Engineering, Huazhong University of Science and Technology, Wuhan, China, in 2014, and the Ph.D. degree in communication and information systems from the School of Computer Science, Wuhan University, Wuhan, in 2020.

He is currently an Associate Professor with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan. His research interests include video coding, image processing, multimedia data processing, and remote sensing.



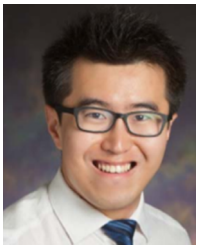
**Yi Xiao** (Member, IEEE) received the B.S. degree from the School of Mathematics and Physics, China University of Geosciences, Wuhan, China, in 2020, and the M.S. degree from Wuhan University, Wuhan, in 2022.

He is currently a Professor with the School of Computer Science and Artificial Intelligence, Zhengzhou University, Zhengzhou, China, and also with the Engineering Research Center of Intelligent Swarm Systems (Ministry of Education) and the National Supercomputing Center. His research interests include remote sensing image/video processing and computer vision.



**Wenxin Huang** (Member, IEEE) received the B.S. degree in information technology from Metropolia University of Applied Sciences, Helsinki, Finland, in 2010, the M.S. degree in geoinformatics from Aalto University, Helsinki, in 2012, and the Ph.D. degree in computer science from Wuhan University, Wuhan, China, in 2020.

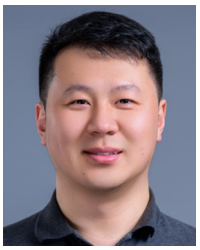
She is currently an Associate Professor with Hubei University, Wuhan, and a Research Officer with the Centre for Frontier AI Research, A*STAR, Singapore. Her research interests include multimedia content analysis and retrieval, large-scale multimedia data mining, and artificial intelligence.



**Bihan Wen** (Senior Member, IEEE) received the B.Eng. degree in electrical and electronic engineering from Nanyang Technological University (NTU), Singapore, in 2012, and the M.S. and Ph.D. degrees in electrical and computer engineering from the University of Illinois at Urbana–Champaign (UIUC), Champaign, IL, USA, in 2015 and 2018, respectively.

He is currently a Nanyang Assistant Professor with the School of Electrical and Electronic Engineering, NTU. His research interests include machine learning, computational imaging, computer vision, image processing, and AI security.

Dr. Wen received the 2022 Early Career Teaching Excellence Award and the 2021 Inspirational Mentor for Koh Boon Hwee Award from NTU. He has also received several best-paper awards and recognitions and has been listed among Stanford University's World's Top 2% Scientists in Artificial Intelligence in 2021 and 2023. He serves as an Associate Editor of IEEE TRANSACTIONS ON CIRCUITS AND SYSTEMS FOR VIDEO TECHNOLOGY and a Guest Editor for *IEEE Signal Processing Magazine* and IEEE JOURNAL OF SELECTED TOPICS IN SIGNAL PROCESSING.



**Xian Zhong** (Senior Member, IEEE) received the B.S. degree in computer science from Wuhan University, Wuhan, China, in 2007, and the Ph.D. degree in computer science from Huazhong University of Science and Technology, Wuhan, in 2013.

He is currently a Professor with the School of Computer Science and Artificial Intelligence, Wuhan University of Technology, Wuhan. He was a Visiting Scholar at Peking University, Beijing, China, from 2021 to 2022, and at ROSE@EEE, Nanyang Technological University, Singapore, in 2024. His research interests include multimodal information processing, neuromorphic vision, and intelligent transportation systems.

Dr. Zhong serves as the Area Chair for IJCAI, IEEE ICASSP, IEEE ICME, and IEEE ICIP. He is an Associate Editor for *IEEE Transactions on Circuits and Systems for Video Technology* and TVC.

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